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Enhancement of high-solid chicken manure anaerobic digestion: Insights into ammonia air stripping pretreatment for microbial resilience and hydraulic retention stabilization

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ABSTRACT

Ammonia inhibition remains a critical barrier in high-solid anaerobic digestion (AD) of nitrogen-rich chicken manure. This study demonstrates an ex-situ ammonia air stripping pretreatment system integrated with semi-continuous AD, achieving dual objectives of nitrogen mitigation and inorganic solids removal. Through 90-day operation at t an organic loading rate (OLR) of 3.1 g-COD/(L·d), pretreated chicken manure (PCM) exhibited a 20 % higher methane yield (225 vs. 187 mL/g-COD) compared to untreated chicken manure (CM). The PCM reactor maintained stable hydraulic retention time (HRT) at 30 days, while the CM reactor suffered a 20 % effective volume loss. Metagenomic analysis revealed stripping-induced microbial selection favoring hydrogenotrophic methanogens (*Methanobrevibacter*) and syntrophic acetate-oxidizers (*Syntrophaceticus*), enabling stable operation at 5.6 g/L total ammonia nitrogen (TAN) versus the inhibitory threshold (6.4 g/L TAN) observed in CM. The pretreatment achieved 28 % total solids (TS) reduction via inorganic solids removal and a 4.5-fold volatile fatty acids (VFAs) surge (39.9 g-COD/L) through microaeration-enhanced hydrolysis. Spearman correlation identified TAN and HRT as dominant determinants of methane yield. This work establishes air stripping as a scalable pretreatment strategy, resolving both biological (ammonia inhibition) and engineering (HRT instability) bottlenecks in chicken manure valorization and bioenergy production.

1. Introduction

The sustainable valorization of agricultural waste is critical for achieving circular bioeconomy objectives. As the world's largest poultry producer, China raised approximately 5.2 billion chickens in 2021, generating over 5 million tons of nitrogen-equivalent manure annually [1]. This resource contains substantial biodegradable organics (4 %–7 % total nitrogen by weight), suitable for energy recovery via anaerobic digestion (AD) [2,3]. However, ammonia inhibition constitutes a fundamental bottleneck in chicken manure mono-digestion at high total solids (TS >10 %). Elevated total ammonia nitrogen (TAN) concentrations consistently exceed 4 g/L, resulting in free ammonia levels >0.5 g/L. This severely disrupts metabolic pathways, reduces methanogenic populations, accumulates volatile fatty acids (VFAs), and causes pH disturbances. Ultimately, this cascade drives >30 % methane yield reductions, chronic reactor instability, and significant economic losses at high organic loading rates (OLR) [4,5], making ammonia inhibition mitigation essential for stable high-solids AD operation.

The root challenge lies in ammonia partitioning dynamics. As TS increases in high-solid systems, free ammonia concentration rises disproportionately due to thermodynamic equilibrium shifts—intensifying inhibition beyond that observed in wet digestion

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[6]. Bi et al. demonstrated that dry AD of chicken manure (20 % TS, OLR = 7.1 g-VS/(L·d)) led to process failure despite relative ammonia tolerance in *Methanosarcina* and *Methanoplasma*; whereas wet systems (5 % TS, OLR = 1.7 g-VS/(L·d)) achieved 280 mL/g-VS methane yield [7]. Crucially, whether high-solid AD can function stably at lowered OLRs under ammonia stress remains unresolved.

Current mitigation strategies exhibit limitations. Biochemical methods (trace element supplementation, *Methanoculleus* spp. bioaugmentation) recover only 15 %–20 % methane under moderate TAN (<4 g/L) but fail in high-solids systems where ammonia partitioning intensifies [8,9]. Co-digestion with high carbon-to-nitrogen (C/N) ratio substrates (e.g., crop residues) reduces TAN by ~34 % through nitrogen dilution [10], yet feasibility is constrained by geographical scarcity of carbon-rich feedstocks near large Chinese biogas plants [11]. Physicochemical TAN removal (dilution, vacuum stripping, precipitation) suffers from prohibitive costs and scalability challenges [12,13].

Novel pretreatments show promise for enhancing AD resilience. Static magnetic fields (SMF), for instance, modulate microbial communities: low-intensity SMF (20 mT) increased sewage sludge biomethane yield by 12.7 %, while high-intensity SMF (1.5 T) inhibited production by 15.1 % [14]. Such targeted interventions underscore pretreatment value. Ammonia stripping leverages NH_4^+/NH_3 phase transitions for nitrogen removal. While established in wastewater treatment [15], its application to high-solid AD faces critical knowledge gaps including: (1) microbial community resilience post-stripping, particularly hydrogenotrophic methanogens' capacity to maintain syntrophy with acetate-oxidizing bacteria under residual ammonia stress; and (2) uncharacterized synergies between ammonia removal and inorganic solids reduction (e.g., grit, sand) [16].

Recent advances confirm that TAN reduction through stripping alleviates acetoclastic methanogens inhibition [17], while introduced air enhances organic solubilization under microaerobic conditions [18]. Bi et al. demonstrated that in-situ biogas stripping reduced TAN by 20 % during high-solid chicken manure AD (15 % TS) [19]. However, no studies have systematically evaluated ex-situ air stripping pretreatment for continuous high-solid chicken manure AD (≥10 % TS), leaving impacts on nitrogen removal, methane production, microbial ecology, and inorganic solids mitigation uncharacterized.

This study pioneers the integration of ex-situ ammonia stripping pretreatment with high-solid chicken manure AD (12 % TS), addressing nitrogen removal efficiency and methane production, microbial community shifts underpinning process stability, and the synergistic effects of ammonia stripping and inorganic solids removal on hydraulic retention time (HRT) preservation. By integrating physicochemical and metagenomic analyses, this work provides novel insights into ammonia stripping scalability for biogas systems, advancing sustainable valorization strategies for nitrogen-rich agricultural waste.

2. Materials and methods

2.1. Inoculum and substrates

The inoculum was obtained from a laboratory-scale continuously stirred tank reactor (CSTR) acclimated to chicken manure digestion for over 6 months at 37 °C (pH = 7.4). The inoculum had a TS content of 4.9 % (wet weight). Volatile solids (VS) comprised 53.1 % of the TS. Fresh chicken manure (TS = 42.1 %, VS/TS ratio = 69.6 %) was collected from a poultry farm in Daqing, China. The manure was diluted to 12 % TS with deionized water, and homogenized using a blender (L12-Y3, Joyoung, China) to ensure uniformity. The substrate was aliquoted into 500 mL vials and stored at 4 °C throughout the experimental period.

2.2. Experimental set-up and operations

Two parallel 10-L CSTRs were operated at 37 °C with a working volume of 8 L. Each reactor was equipped with a paddle-type agitator (100 rpm), a water jacket, and a heated circulation system for temperature control. Both reactors were fed daily with either pretreated chicken manure (PCM) or untreated chicken manure (CM) at an OLR of 3.1 g-chemical oxygen demand (COD)/(L·d) and an HRT of 30 days, reflecting industrial-scale biogas plant conditions [11]. Biogas production, composition (CH_4 and CO_2), and pH levels were monitored daily. Additionally, the concentrations of COD, TAN, and VFAs in the digestate were analyzed every 10 days.

The ammonia stripping system (Fig. 1) was integrated into the PCM reactor, operating at 37 °C with a 2-day HRT. Key parameters included cyclic aeration (15 s every 5 min at 2 L/min flow rate) following prior optimization [19]. The stripped ammonia was then absorbed in an 800 mL scrubber containing 5 M sulfuric acid (H_2SO_4) for neutralization, which facilitated the recovery of ammonium sulfate. The CM reactor operated identically without stripping.

2.3. Chemical analysis

The determination of TS and VS was performed using gravimetric analysis. Samples were dried at 105 °C for a minimum of 24 h and at 600 °C for at least 6 h. The concentrations of total COD, soluble COD, and TAN were measured following the methods outlined by the American Public Health Association [20]. VFAs including acetate, propionate, butyrate, valerate, and caproate were analyzed according to a previous study [21]. pH was measured using a pH meter (PHS-3C; Yidi, Shanghai, China). Biogas yield was quantified daily with a wet biogas meter (LML-1; Hongke, Shanghai, China) and calibrated to standard conditions of temperature (273.15 K) and pressure (1 atm). Biogas composition was analyzed using a gas chromatograph (GC-7890; Jinpu, Shandong, China) equipped with a 2 m long and 3 mm inner diameter capillary column (TDX-01; Jinpu). The column, injector, and detector temperatures were maintained at 100 °C, 100 °C, and 50 °C, respectively, with hydrogen serving as the carrier gas.

2.4. Microbial community analysis

Digestate samples (PCM and CM reactors) and substrates were collected on day 83 (steady-state) and stored at −80 °C until further metagenomic analysis. Triplicate samples were sent to Meiji Company for DNA extraction, quality testing, sequencing, and sequence analysis. Specifically, DNA was extracted using the CTAB/SDS method, followed by purification (1 % agarose gel) and quantification (NanoDrop 2000, Thermo Fisher). The purified DNA was used for PCR amplification of bacterial and archaeal 16S rRNA genes. For bacterial sequences, the primers was 338F (5′-ACTCCTACGGGAGGCAGCAG-3′) and 806R (5′-GGACTACHVGGG TWTCTAAT-3′) were used, while for archaeal sequences, the primers 524F (5′-TGY CAG CCG CCGCGGTAA-3′) and 958R (5′-YCCGGCGTTGAVTCCAATT-3′) [22]. The PCR products were purified, quantified, and sequenced using the Illumina MiSeq platform (2 × 300 bp paired-end). Raw sequences were quality-filtered (Trimmomatic v0.38), denoised (DADA2), and clustered into operational taxonomic units (OTUs) at 97 % similarity (UPARSE) [19]. Taxonomic classification was performed using the SILVA database (v138), and functional annotation utilized FAPROTAX [23]. Most annotated species were assigned to groups associated with essential metabolic and ecological functions, including C, N, and S biogeochemical cycling and organic matter degradation.

2.5. Calculations and statistics

Data are presented as mean ± standard deviation ($n = 3$). Significance between reactors was assessed via Student's t-test ($p < 0.05$). All analyses were performed in Origin 9.1.

Hydrolysis, acidogenesis, and acetogenesis efficiencies were calculated using COD mass balance equations (Eqs. (1)–(3)) [24].

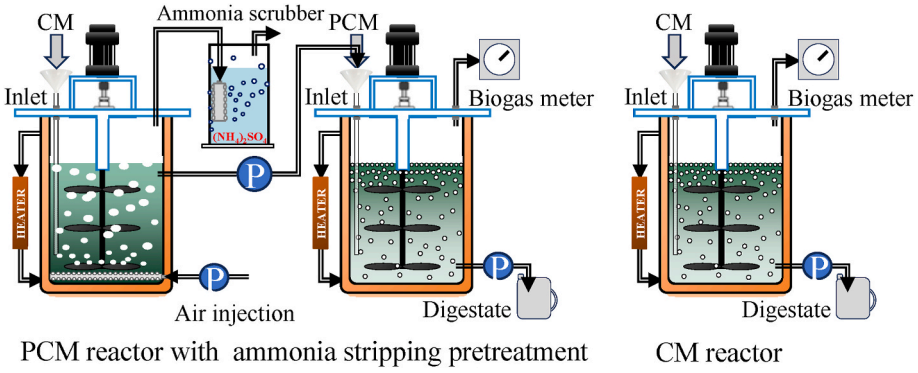


Fig. 1. Schematic diagram of the experimental setup for the anaerobic digestion process. The diagram illustrates the key components of the system, including the ex-situ ammonia air stripping unit and the semi-continuous anaerobic digester.

$$\text{Hydrolysis}\ (\%) = \frac{\text{SCOD} - \text{SCOD}_{\text{CM}}}{\text{TCOD}_{\text{CM}} - \text{SCOD}_{\text{CM}}} \times 100 \quad (1)$$

$$\text{Acidogenesis}\ (\%) = \frac{\text{COD}_{\text{VFA}} - \text{COD}_{\text{VFACM}}}{\text{TCOD}_{\text{CM}} - \text{COD}_{\text{VFACM}}} \times 100 \quad (2)$$

$$\text{Acetogenesis}\ (\%) = \frac{\text{COD}_{\text{Acetate}} - \text{COD}_{\text{AcetateCM}}}{\text{TCOD}_{\text{CM}} - \text{COD}_{\text{AcetateCM}}} \times 100 \quad (3)$$

where $TCOD_{CM}$ and $SCOD_{CM}$ represent the total and soluble COD in the CM, respectively; and $SCOD$ represents the soluble COD in the PCM. The COD of the VFAs was determined by calculating the oxygen demand of each VFA and represents the total VFA concentration. COD_{VFA} and $COD_{Acetate}$ refer to the COD in the PCM. COD_{VFACM} and $COD_{AcetateCM}$ indicate the COD in the CM.

3. Results

3.1. Substrate transformation through pretreatment process

Stripping pretreatment significantly altered the physicochemical properties of chicken manure (Table 1). TS decreased by 28 % ($p < 0.05$), indicating efficient inorganic removal (e.g., ash), while total COD remained stable ($p = 0.766$), confirming minimal organic matter loss—a critical factor for preserving energy potential. Soluble COD increased by 51 % (from 36.1 to 54.4 g-COD/L, $p < 0.05$), reflecting enhanced solubilization of particulate organics. Additionally, VFAs surged 4.5-fold (from 8.9 to 39.9 g-COD/L, $p < 0.05$), predominantly consisting of acetate (30 %), propionate (18 %), and butyrate (20 %) (Table 1), suggesting that microaeration promotes hydrolysis and acidogenesis, consistent with studies on complex wastes [18].

In terms of nitrogen dynamics, total nitrogen decreased by 14 % (from 6.65 to 5.72 g/L, $p < 0.05$), while TAN increased by 108 % (from 2.4 to 5.0 g/L, $p < 0.05$) (Table 1). This paradox arises from the dual mechanisms of pretreatment: the volatilization of free NH_3 and the hydrolysis of organic nitrogen (e.g., proteins) into NH_4^+, which tends to accumulate unless fully stripped.

Conversion efficiencies for hydrolysis (31.9 %), acidogenesis (36.6 %), and acetogenesis (12.5 %) aligned closely with thermophilic anaerobic benchmarks (70 °C) [25], demonstrating that microaerobic stripping can achieve comparable efficiency at lower mesophilic temperatures (37 °C) with reduced energy costs.

Table 1
Characteristics of the chicken manure and pretreated chicken manure.

Items	Unit	Chicken manure	Pretreated chicken manure	Inoculum
TS	g/L	112.4 ± 3.2	85.8 ± 2.5	6.8 ± 1.8
pH		7.85 ± 0.13	6.83 ± 0.15	7.35 ± 0.13
TN	g/L	6.65 ± 0.47	5.72 ± 0.52	ND
TAN	g/L	2.39 ± 0.38	4.97 ± 0.24	4.93 ± 0.22
TCOD	g/L	93.51 ± 0.72	91.77 ± 0.79	ND
SCOD	g/L	36.14 ± 0.12	54.43 ± 0.57	ND
Acetate	g COD/L	5.30 ± 0.06	16.33 ± 0.48	ND
Propionate	g COD/L	1.90 ± 0.02	9.77 ± 0.02	ND
Isobutyrate	g COD/L	0.05 ± 0.02	0.26 ± 0.02	ND
Butyrate	g COD/L	1.37 ± 0.02	10.76 ± 0.1	ND
Isovalerate	g COD/L	0.10 ± 0.02	0.37 ± 0.02	ND
Valerate	g COD/L	0.02 ± 0.02	1.18 ± 0.02	ND
Hexanoate	g COD/L	0.16 ± 0.02	1.24 ± 0.02	ND
VFAs	g COD/L	8.90 ± 0.18	39.90 ± 0.25	ND

Averaged values are presented ± standard deviation; TS = total solids; TN = total nitrogen; TAN = total ammonium nitrogen; TCOD = total chemical oxygen demand; SCOD = soluble chemical oxygen demand; VFAs = volatile fatty acids; ND = no data.

3.2. The enhancement of methanogenic performance

Throughout the 87-day fermentation, the PCM reactor maintained lower TAN levels (Fig. S1) and achieved higher methane productivity than the untreated CM reactor (Fig. 2). Notably, methane yield in the PCM reactor remained stable (linear regression slope, $k = -0.04$), whereas the CM reactor exhibited a significant decline ($k = -0.45$), decreasing from 203 to 187 mL/g-COD across three HRT stages (Fig. 2a). By the third HRT, the PCM reactor achieved a yield of 225 mL/g-COD (64 % of the theoretical maximum of 350 mL/g-COD), outperforming the CM reactor by 20 % ($p < 0.05$, Fig. 2b).

Spearman's correlation analysis (Fig. 3a) of operational parameters (Table 2) revealed methane yield was negatively correlated with TAN, OLR, VFA accumulation, and feed concentration (TS and COD), but positively correlated with HRT and substrate pretreatment. Under suitable conditions (OLR = 3.1 g-COD/(L·d), HRT = 30 d), PCM achieved 225 mL/g-COD with low VFAs (1.9 g/L) and moderate TAN (5.6 g/L). In contrast, CM yielded only 187 mL/g-COD despite identical parameters, accompanied by higher TAN (6.4 g/L) and VFAs (6.5 g/L). PCM surpassed literature benchmarks at similar or lower OLRs [26–28], primarily due to ammonia stripping pretreatment, which reduced TAN by 12.5 % versus CM ($p < 0.05$; Table 2).

The pretreatment system also removed sand and stones from the manure, ensuring sufficient HRT within the PCM reactor. Industrial biogas plants prioritize stable operation under high OLR and low HRT for profitability [32]. Unscreened chicken manure requires a minimum HRT of 40 days [30] due to 8 %–10 % TS as sand amd stones entrapped

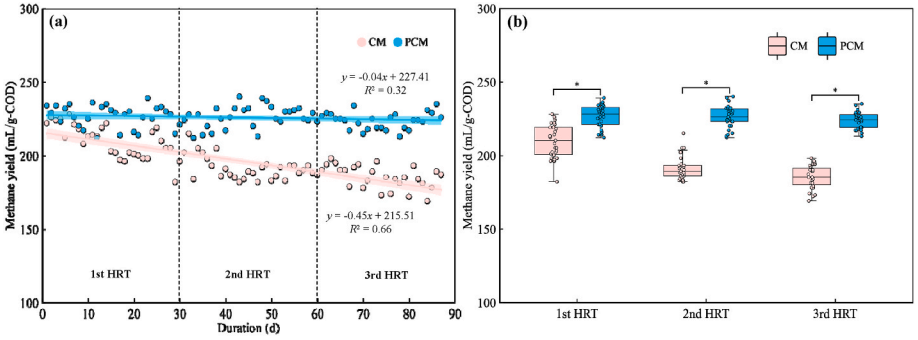


Fig. 2. Methane production during 87 days operations with three hydraulic retention time. (a) Methane production trends over a period of 87 days of operation and analyzed by liner fitting; (b) compares methane yields across three distinct HRT.

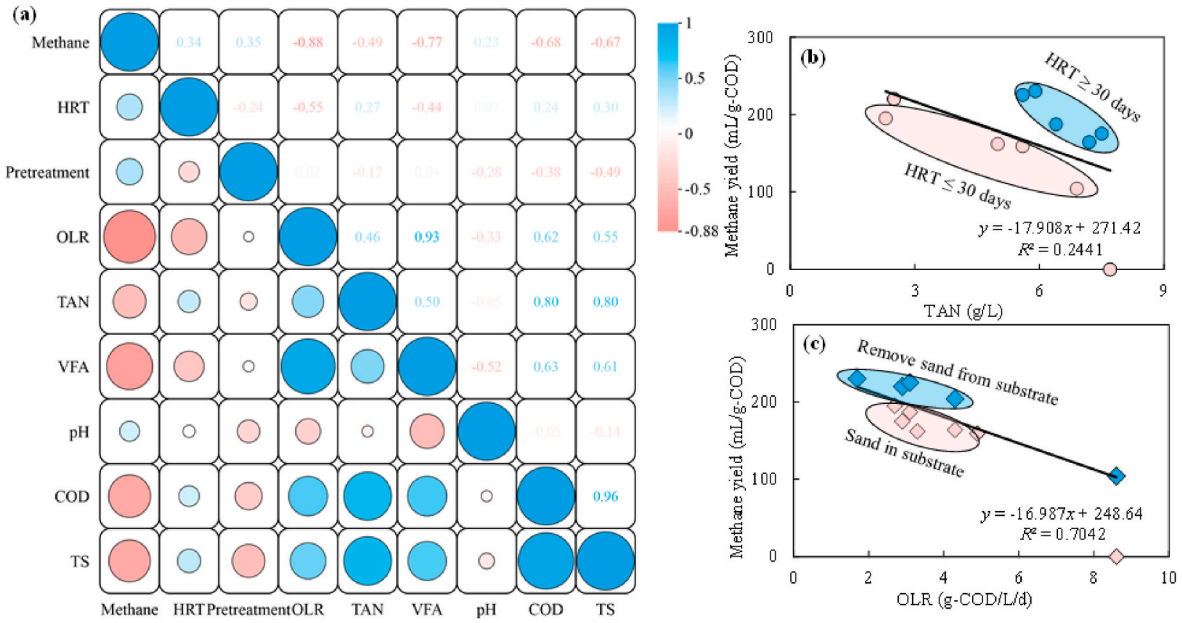


Fig. 3. Comprehensive analysis of key operational parameters influencing mesophilic anaerobic digestion of chicken manure. (a) Spearman rank correlation coefficients between critical parameters, including methane yield, HRT, pretreatment, OLR, TAN, VFA, pH, feed COD, and TS. This correlation analysis helps identify relationships and potential interdependencies among operational factors; (b) the effects of TAN and HRT on methane yield; (c) examines the relationship between OLR and pretreatment methods with methane production. This detailed analysis is based on the data presented in Table 2, aiming to elucidate the operational dynamics affecting the digestion process.

Table 2
Comparison of CSTR performance during the mesophilic anaerobic digestion of chicken manure.

Feed COD (g-COD/L)	Feed TS (g-TS/L)	Pretreatment	HRT (Days)	OLR (g-COD/(L·d))	TAN (g/L)	VFAs (g/L)	Methane yield (mL/g-COD)	pH	References
80	89	No	30	2.7	2.3	0.5	195	8.1	[26]
100	105	No	30	3.3	5	1.5	162	8.2	[26]
57	50	Screened	20	2.9	2.5	0.3	220	8.3	[27]
87	76	Screened	20	4.3	5.6	3.1	204	8.3	[27]
99	103	Screened	20	4.9	5.6	5.7	159	8.2	[28]
99	103	Screened	60	1.7	5.9	0.3	230	7.8	[28]
171	150	Screened	20	8.6	6.9	22.3	104	7.8	[29]
171	150	No	40	4.3	7.2	6.4	164	8.3	[30]
171	150	No	60	2.9	7.5	0.5	175	8.5	[30]
171	150	No	20	8.6	7.7	14.8	0	8.1	[31]
94	112	No	30	3.1	6.4	6.5	187	8.1	This study
92	86	Ammonia stripping	30	3.1	5.6	1.9	225	8.2	This study

COD = chemical oxygen demand; HRT = hydraulic retention time; OLR = organic loading rate; TAN = total ammonium nitrogen; VFAs = volatile fatty acids.

in fecal matter [4,30]. Screened manure operates reliably at a HRT of 20 days and an OLR of 5.0 g/(L·d) [7].

Linear regression was conducted to assess the effects of TAN and HRT on methane yield. The analysis confirmed that increased HRT (≥30 days) enhanced methane yield, with higher HRT values exceeding the fitted trendline and lower values falling below predictions (Fig. 3b). PCM consistently exceeded regression predictions, while untreated CM fell below the trendline, indicating inorganic removal improved methane production (Fig. 3c). The CM reactor's effective volume decreased by approximately 20 % during digestion (actual HRT = 24 days), directly reducing methane output. Conversely, PCM maintained stable productionat the target HRT of 30 days.

These results underscore the dual role of ammonia stripping: (1) reducing TAN by 12.5 %, alleviating inhibition of acetoclastic methanogens (e.g., *Methanosaeta*); and (2) removing inorganic solids, preventing HRT shortening via solids accumulation. This highlights the stripping efficiency in enabling stable high-rate digestion (HRT = 30 days) without sacrificing methane output—a critical advantage for industrial biogas plants prioritizing reactor volume minimization and operational continuity [32]. Furthermore, the removal of sand and stones not only preserved HRT but also enhanced process economics by reducing maintenance costs associated with pump wear and grit removal [33].

3.3. The enhancement of methane conversion rate

Comparing the conversion of organic matter into soluble COD, VFAs, and methane in PCM and CM reactors (Fig. 4), the CM reactor exhibited a higher fraction of particulate COD, indicating a higher proportion of non-degraded organic matter (28.9 %) than that in the PCM reactor (23.9 %). This result suggested that stripping pretreatment enhanced the efficiency of organic matter breakdown in the PCM reactor. In addition, the CM reactor exhibited higher levels of soluble COD and VFAs than the PCM reactor, likely due to TAN inhibition of methanogenesis, leading to VFA accumulation [7,34,35]. The PCM reactor demonstrated that methane conversion rate of 65.7 %, which was 23.1 % higher than that of CM reactor, highlighting the benefits of stripping pretreatment in enhancing hydrolysis, acidogenesis, and methane production.

3.4. Variations of bacterial community

High-throughput sequencing revealed Firmicutes (66.4 %–83.1 %) as the dominant bacterial phylum across all samples (Fig. 5a), consistent with studies linking its hydrolytic and acidogenic versatility to enhanced VFA production under high OLRs conditions [34,36,37]. Substantial variations among the sampled specimens were observed at the genus level (Fig. 5b). The higher abundance of *Lachnospiraceae* NK3A20 in PCM versus untreated CM aligns with accelerated carbohydrate degradation [38], explaining the 4.5-fold VFA surge during pretreatment (Table 1). The PCM reactor was dominated by *Fastidiosipila*, *Gallicola*, and *Lachnospiraceae NK3A20*, contrasting sharply with CM reactor's prevalence of *Proteiniphilum*, MBA03, and *Thiopseudomonas* in the CM reactor. This enrichment of specialized genera (*Fastidiosipila*, *Gallicola*) capable of converting uric acid and proteins to acetate and butyrate [19,39] diverges from CM reactor's nitrogen-tolerant genera (*Proteiniphilum*, *Thiopseudomonas*) thriving under methanogenic inhibition. Elevated VFAs generated substrate pools for syntrophic consortia. Consequently, Clostridium and Syntrophaceticus were slightly enriched in PCM (1.1 % vs. 0.7 % in CM), potentially aiding methane production [40].

Functional annotation using FAPROTAX identified 43 ecological functions, with chemoheterotrophy and fermentation (20 %–50 %) being predominant (Fig. 5c). The pathogen-related functions associated with animal parasites or symbionts, human gut, mammalian gut, and human pathogens were detected only in untreated manure, suggesting efficient pathogen removal during stripping pretreatment and AD processes. Furthermore, ecological functions related to nitrite respiration, reduction, and ammonification were exclusively detected at high levels in untreated manure, indicating the presence of nitrogen-metabolizing bacteria. These results suggested that the pretreatment process reduced the nitrogen content of the substrate.

3.5. Variations of archaeal community

The Archaeal communities in unpretreated CM, PCM and reactors were compared at the phylum level. Euryarchaeota dominated archaeal communities except in untreated CM, where Crenarchaeota prevailed (Fig. 6a). Euryarchaeota includes key AD methanogens. Significant genus-level differences existed between the CM and PCM reactors (Fig. 6b). *Methanobrevibacter* was enriched in the PCM reactor, contrasting with *Methanobacterium* and *Methanoculleus* in the CM reactor. Previous studies have shown that methanogens in these genera are commonly found in AD processes with high levels of TAN and have demonstrated tolerance to ammonia, effectively acting as methanogenic partners in syntrophic acetate oxidation [7,41]. *Methanobacterium* and *Methanoculleus* were both identified in the anaerobic reactors used to treat chicken manure, and their dominance was observed in conjunction with significant VFA accumulation and low methane production. *Methanobrevibacter* enrichment in PCM reactor, known for ammonia tolerance and syntrophic acetate oxidation, likely contributed to superior methane production [42].

Functional annotation of the archaeal taxa was performed using the FAPROTAX database and the archaeal communities were annotated with 10 ecological functions, including hydrogenotrophic methanogenesis (mainly in PCM, CM and PCM reactors) and aerobic ammonia oxidation in the untreated CM samples (Fig. 6c).

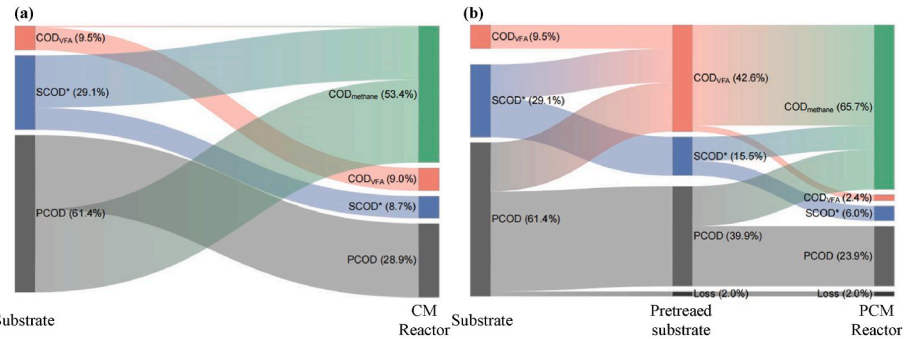


Fig. 4. Mass balance based on COD in CM reactor (a) and PCM reactor (b).. (Note: TCOD and SCOD represent the total and soluble COD in the substrate/digestatate; COD_{VFA} represent total VFA concentration in the substrate/digestatate; PCOD represents particulate COD in the substrate/digestatate; $COD_{methane}$ represents the COD transformed to methane; SCOD* = SCOD- COD_{VFA}, and TCOD = PCOD + SCOD* + COD_{VFA} + $COD_{methane}$.)

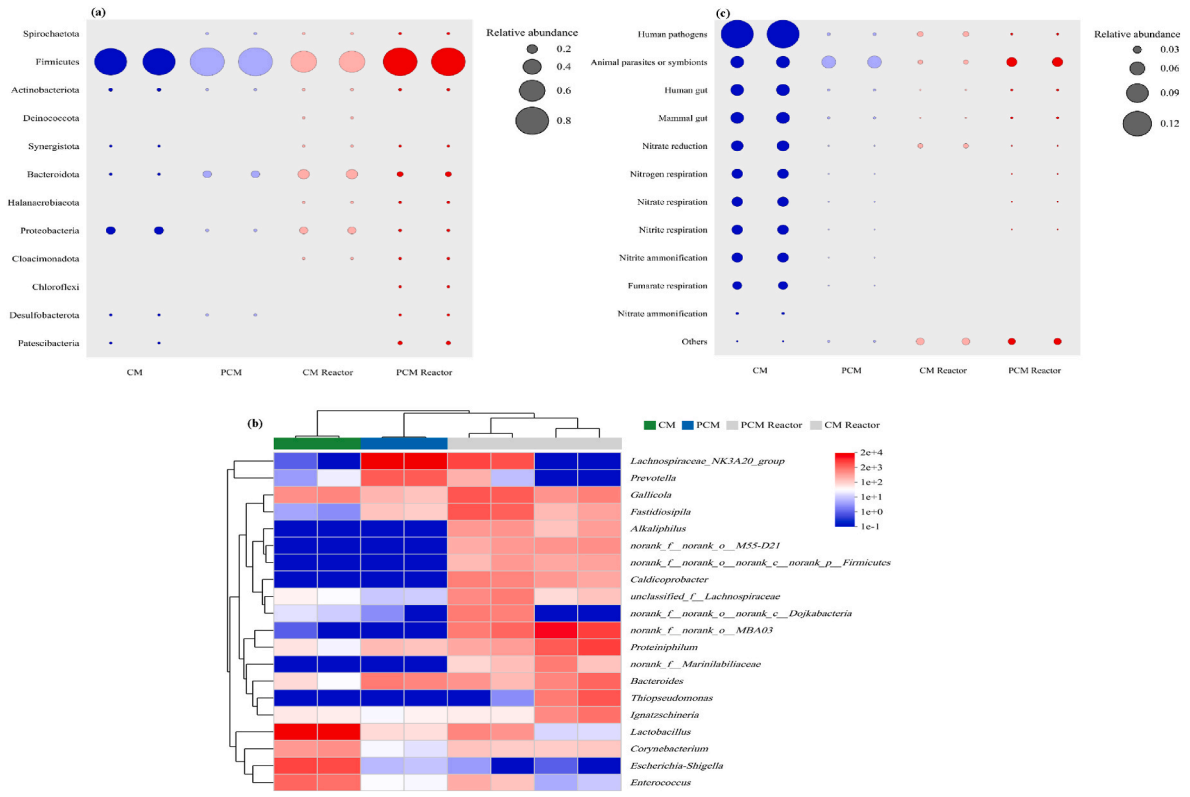


Fig. 5. The effect of pretreatment on the composition and functionality of bacterial community in anaerobic digestions of chicken manure. (a) Relative abundance of the bacterial relative abundance at the phylum level; (b) hierarchical cluster analysis of interesting bacteria archaea; and (c) functional annotation of bacterial taxa using the FAPROTAX database. (Note: CM, chicken manure; PCM, pretreated chicken manure).

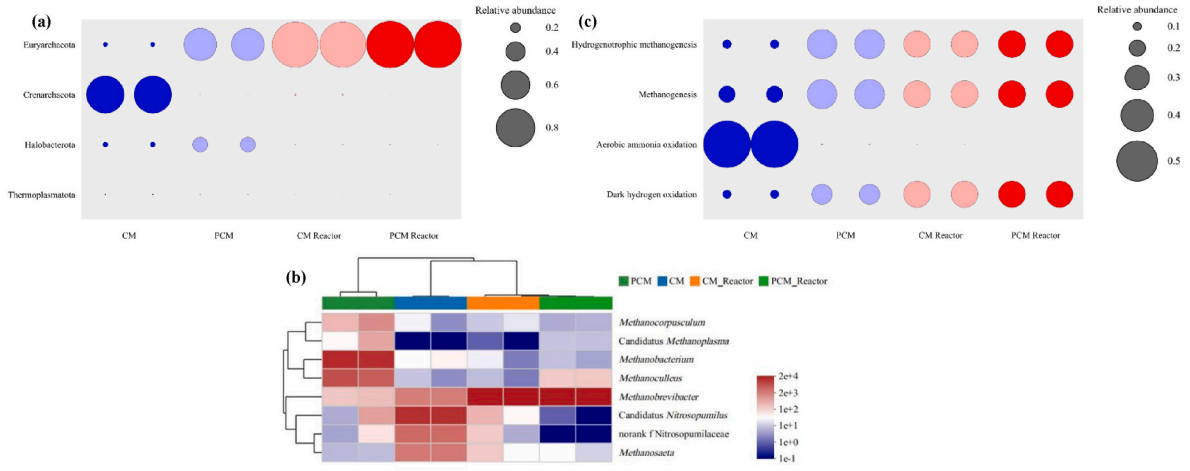


Fig. 6. The effect of pretreatment on the composition and functionality of archaeal community in anaerobic digestion of chicken manure. (a) Archaeal relative abundance at the phylum level; (b) hierarchical cluster analysis of interesting archaea; and (c) functional annotation of archaeal taxa using the FAPROTAX database. (Note: CM, chicken manure; PCM, pretreated chicken manure).

4. Discussion

4.1. Comparison with other similar studies

This study establishes ex-situ ammonia air stripping as a highly effective pretreatment strategy for overcoming ammonia inhibition and HRT instability in high-solids AD of chicken manure. By integrating stripping with semi-continuous AD (PCM reactor), a significant 20 % increase in methane yield (225 vs. 187 mL/g-COD) was achieved while maintaining stable HRT of 30 days. This was underpinned by a 28 % reduction in TS through inorganic solids removal and a dramatic 4.5-fold increase in VFAs (39.9 g-COD/L) facilitated by microaeration-enhanced hydrolysis. Metagenomic analysis revealed this performance was driven by microbial community restructuring, specifically the selective enrichment of hydrogenotrophic methanogens (*Methanobrevibacter*) and syntrophic acetate-oxidizing bacteria (*Syntrophaceticus*), enabling stable methanogenesis at elevated TAN levels (5.6 g/L). Fig. 7 summarizes the proposed process mechanism for enhanced AD performance.

These findings align with, yet significantly advance, previous

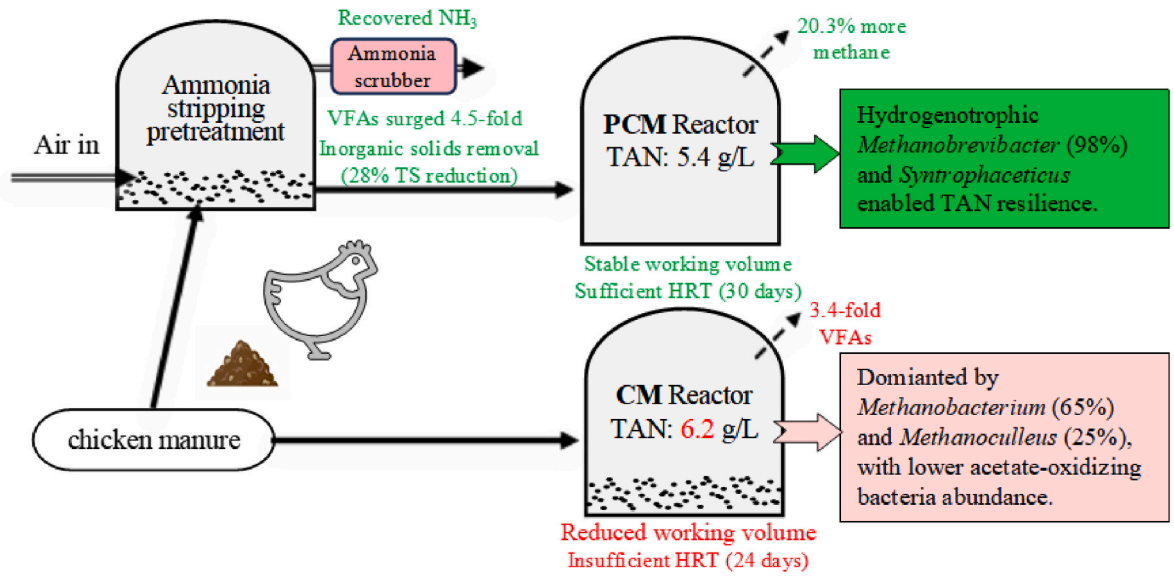


Fig. 7. The proposed process mechanism for enhanced AD performance via ammonia air stripping pretreatment.

research on ammonia stripping for nitrogen-rich wastes. The demonstrated efficacy in mitigating ammonia toxicity confirms core principles established in studies across different feedstocks. For instance, stripping of piggery wastewater significantly enhanced methane productivity [43], while studies on poultry manure also reported substantial increases in methane yield (e.g., 30–34 % improvements [19,44]) through ammonia removal. Our observed shift towards hydrogenotrophic methanogenesis, supported by enriched *Syntrophaceticus* and *Methanobrevibacter*, resonates with findings where stripping or similar mitigations (e.g., biochar addition [45]) enriched hydrogenotrophic pathways (*Methanoculleus*, *Methanothermobacter*) or syntrophic partners under high ammonia stress [46–48]. Furthermore, this study directly addresses a critical operational bottleneck often overlooked in fundamental research: HRT instability caused by inorganic solids accumulation. The removal of grit and sand during stripping proved essential for preventing reactor clogging and preserving stable HRT, addressing a key failure mode in industrial-scale systems [19,44]. This operational benefit, combined with the biological mitigation, represents a holistic engineering solution. While variations exist in methane yield improvements reported (ranging from ~20 % to >350 % depending on initial inhibition severity and mitigation strategy [44,48,49]), the magnitude of our increase (20 %) is consistent within the context of successful pretreatment under high-solids conditions and aligns with optimized systems like those enhanced by biogas recirculation stripping [13]. The differences can be attributed to factors such as pretreatment intensity (pH, duration, microaeration inclusion), stripping configuration (ex-situ vs. in-situ [13,50]), substrate properties, and operating conditions (OLR, temperature). Our focus on microaeration-enhanced hydrolysis during stripping as a driver for VFA surge adds a novel operational dimension to the pretreatment effect. The concomitant reduction in pathogen-associated functions adds another valuable benefit consistent with improved process sanitation observed in other stripping studies [50].

4.2. Research limitations and future perspectives

While this study demonstrates the clear benefits of ex-situ ammonia air stripping integrated with AD for high-solids chicken manure management, contributing significantly to both biological stability and operational robustness, several limitations warrant consideration.

As highlighted for future work, the optimization of stripping duration and gas-liquid contact efficiency was identified as crucial but not exhaustively explored here. Maximizing net nitrogen removal while minimizing organic carbon loss remains a critical engineering trade-off [43]. Findings confirmed the importance of pH control, but potential sodium ion inhibition at very high pH values (>10–11), as noted in wastewater stripping studies [43,46], requires careful management in manure systems where large alkali additions might be needed, suggesting low-cost alkaline agents or pH control strategies are worth further investigation.

The integration of ex-situ stripping adds complexity compared to single-stage AD or simpler additive strategies (e.g., biochar, allophane [45,48,49]). While proven effective here, the economic viability and energy balance of this two-stage approach (including the microaeration step) at full industrial scale require detailed techno-economic analysis and life cycle assessment, building on pilot-scale validations like those initiated for related technologies [50].

Although metagenomic analysis revealed key shifts (Methanobrevibacter, Syntrophaceticus enrichment), the long-term microbial community resilience and functional stability under varying substrate loads or process disturbances beyond the tested effective volume loss scenario need further longitudinal investigation. The integration of "microbial resilience metrics with process engineering parameters" proposed here provides a framework, but its predictive power needs validation across more diverse conditions. The interplay between stripping-removed nutrients (beyond just TAN) and microbial function was not fully characterized.

The study utilized a specific chicken manure source. The effectiveness and optimal stripping parameters might vary with manure composition (depending on diet, bedding, collection methods), which is common to all AD research on animal wastes. Testing robustness across a range of manure compositions would strengthen broader applicability claims.

5. Conclusions

This study demonstrates that ex-situ ammonia air stripping effectively overcomes ammonia inhibition and HRT instability in high-solids chicken manure digestion. The pretreatment enabled robust methanogenesis by enriching specialized ammonia-tolerant microbial consortia, while inorganic solids removal prevented reactor volume loss. Concurrent ammonia recovery yielded valuable fertilizer, enhancing process sustainability. Although operational parameters warrant further optimization, this integrated strategy resolves critical biological and

engineering bottlenecks. Implementation at scale offers a viable pathway for sustainable agricultural waste valorization and bioenergy recovery.

CRediT authorship contribution statement

Xinhui Yu: Writing – original draft, Software, Data curation. **Fangtong Wei:** Validation, Data curation. **Haipeng Wang:** Writing – review & editing. **Shaojie Bi:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Changjiang Zhao:** Writing – review & editing. **Yanjie Wang:** Writing – review & editing.

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Declaration of competing interest

The authors declare that they have no known competing interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biombioe.2025.108098.

Data availability

Data will be made available on request.

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